Look. Feel.

What magic are user interfaces are made of?

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It all started back in November 2006, when we were coming up with new ideas for the interface for Digit DVD (which, you might remember, made its debut in December 2006). In our (shameless) imitation of the Vista desktop, several questions popped up: why is the Start button on the bottom-left? Who decides where the close button goes, and why must we use these little boxes called "windows", when we maximise them to fit our desktops anyway? But it was 4 am; we were sleep-deprived, and had deadlines.

But the questions still remain. How do interfaces work for us, and why in heaven's name do we actually enjoy some of them?

Let's start with some background...

The Science

Yes, there's actual science at work here, and it dictates both the way user interfaces (UIs) are laid out, and how they become enjoyable.

The Law

All interfaces – more specifically, the ones that require pointing – start with a mathematical model called Fitts's Law (some people will tell you it's Fitt's law, or Fitts' law). They are wrong. The law describes the time to acquire a target (that is, be able to click on a button, hyperlink, or menu item) as a function of the distance to it, and the size of the target. You can test this right now: if your mouse cursor is in the middle of your screen, it'll take you less time to reach an icon that's closer – obviously. If you turn on Large Desktop Icons, you'll be able to hit those icons faster – by milliseconds, of course, but those add up.

There are important conclusions to be drawn here. Firstly, big targets are good. Secondly, targets should be as close to the mouse pointer as possible. The most important conclusion, however, is this: the edges and corners of the screen are infinitely huge.

Think of it this way: no matter how much you move the mouse, the cursor will never go beyond the edge. So unlike a target that's placed in the center of the screen, you'll never overshoot a target that's placed at its edge.

What does all this mean? Ah, but we must keep you on the edge of your seat for a bit longer. There's another aspect to UIs, and it comes from the science of happiness.

The Flow

Since the 80s, Apple has been throwing its weight behind an interface style called Direct Manipulation (DM). With the ability to point and click at objects on screen comes the desire to control them – to move and rotate and push and pull and drag and drop (aha!) – and DM interfaces must fulfil those desires. Interfaces should feel like 'driving a car'.

But why driving?

In 1990, Mihaly Csikszentmihalyi, a Hungarian professor studying happiness, described the concept of flow – a state of mind where you feel immersed, involved, and in control. And one very important source of flow is...yes, driving.

And so, DM interfaces must give you a sense of flow. Games do this all the time – the best games, as you have no doubt experienced, match themselves perfectly to your skills. They never feel so hard that you want to give up, but never so easy that you feel bored. Like driving.

And so, armed with these two pieces of knowledge, we can explain the mysteries of the universe! At least, those that pertain to interfaces.

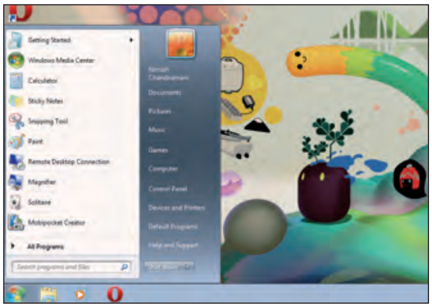
A Mac desktop, with icons optimised for maximum hit-ability

The Answers

And now, we come back to some of our original mysteries, and some new ones.

The case of the Windows Start button

Now that we know the implications of Fitts's law, this one isn't hard to crack:


Even though it's a big glowy round button, the Windows 7 Start button still respects the corner

corners are good, so if you've got something important you want users to access – say, a single button that gives users access to every function on their computer – you stick it in a corner of the screen. You'll notice that even though the big, round start buttons in Windows Vista and 7 don't touch the corner, you can still activate them by just throwing your mouse cursor to the bottom-left and clicking.

The case of the Mac menu bar

This one is a common irritant for Windows users who are used to menu bars being inside their programs – on the Mac, the menu bar is always on the top of the screen. Despite the culture shock, this is a good thing, according to Fitts's law. The top edge of the screen is infinitely high, so no matter how far up you move the mouse cursor, you'll always be able to use the menu bar.